EXPERIMENT 8

MEMORY MANAGEMNET TECHNIQUE-

BEST FIT, FIRST FIT

FIRST FIT

```
#include <stdio.h>
int main(){
    int b[10], p[10], i, j, n, m, idx;

    printf("Enter number of memory blocks: ");
    scanf("%d",&n);

    printf("Enter sizes of memory blocks:\n");
    for(i=0;i<n;i++) scanf("%d",&b[i]);

    printf("Enter number of processes: ");
    scanf("%d",&m);

    printf("Enter sizes of processes:\n");
    for(i=0;i<m;i++) scanf("%d",&p[i]);

    // FIRST FIT
    printf("\n--- First Fit Allocation ---\n");
    int b1[10];
    for(i=0;i<n;i++) b1[i] = b[i];

    for(i=0;i<m;i++){
        for(j=0;j<n;j++){
```

```
        if(b1[j] >= p[i]){
            printf("Process P%d (%d) -> Block B%d\n", i, p[i], j);
            b1[j] -= p[i];
            break;
        }
    }
}

// BEST FIT
printf("\n--- Best Fit Allocation ---\n");
int b2[10];
for(i=0;i<n;i++) b2[i] = b[i];

for(i=0;i<m;i++){
    idx = -1;
    for(j=0;j<n;j++){
        if(b2[j] >= p[i]){
            if(idx == -1 || b2[j] < b2[idx]){
                idx = j;
            }
        }
    }

    if(idx != -1){
        printf("Process P%d (%d) -> Block B%d\n", i, p[i], idx);
        b2[idx] -= p[i];
    }
    else{
        printf("Process P%d (%d) -> Not Allocated\n", i, p[i]);
```

```
        }
    }

    return 0;
}
```

OUTPUT

First Fit Allocation

Process	Size	Block Allocated
P1	212	Block 2 (500)
P2	417	Block 5 (600)
P3	112	Block 2 (288)
P4	426	Not Allocated

Best Fit Allocation

Process	Size	Block Allocated
P1	212	Block 4 (300)
P2	417	Block 2 (500)
P3	112	Block 3 (200)
P4	426	Block 5 (600)